CS-4

20/06/24 M/S

USN | | | | | | | | | | |

[23BSMA21]

SECOND SEMESTER B.Tech EXAMINATION, JUNE - 2024

PROBABILITY AND VECTOR SPACES

Time: 3 Hours **Maximum Marks: 100**

Instructions:

i. Missing data may be suitably assumed.
ii. Statistical tables will be provided

ANSWER ALL QUESTIONS

		PART - A			**05x02=10**
1.	**a.**	A card is drawn randomly from a pack of 52 cards, what is the probability of getting a king or queen?	L1	CO1	**02**
	b.	From the following probability distribution, find the missing probability.	L2	CO2	**02**

x_i	-2	-1	0	1	2
$P(x_i)$	0.2	0.3	0.2	?	0.1

	c.	Define Linearly dependent and independent vectors.	L2	CO3	**02**
	d.	Find the linear transformation $f: R^2 \to R^2$, such that $f(1,0) = (1,1)$ and $f(0,1) = (-1,2)$.	L3	CO4	**02**
	e.	Find the Eigen values of the matrix $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$.	L3	CO6	**02**
		PART - B			**06x05=30**
2.	**a.**	A bag contains 40 tickets numbered 1,2,3, …… 40 of which four are drawn at random and arranged in ascending order ($t_1 < t_2 < t_3 < t_4$). Find the probability of t_3 being 25.	L2	CO1	**05**
		OR			
	b.	In a school 25% of the students failed in first language, 15% of the students failed in second language and 10% of the students failed in both. If a student is selected at random find the probability that **i.** He failed in first language if he had failed in the second language **ii.** He failed in second language if he had failed in the first language **iii.** He failed in either of the two languages.	L2	CO1	**05**
3.	**a.**	If X is Uniform between 8 and 20 with mean 14 and variance 12. Find probability $P\ (\mid X - 6 \mid \leq 15)$ and $P\ (15 \leq X \leq 17)$.	L3	CO2	**05**
		OR			
	b.	The length of telephone conversation in a booth has been an exponential distribution and found on an average to be 5 minutes. Find the probability that a random call made from this booth (i) ends less than 5 minutes (ii) between 5 and 10 minutes.	L3	CO2	**05**
4.	**a.**	Show that the subset $w = \{(x_1, x_2, x_3)/x_1 + x_2 + x_3 = 0\}$ of the vector space $V_3(R)$ is a subspace of $V_3(R)$.	L2	CO3	**05**
		OR			
	b.	Express the vector (3,5,2) as a linear combination of the vector (1,1,0), (2,3,0), (0,0,1) $of\ V_3(R)$.	L2	CO3	**05**

JAIN (Deemed to be University) Learning Resource Centre Jain Global Campus Ramanagara-562112

5.	**a.**	Find the matrix of linear transformation $T: V_3(R) \to V_2(R)$ defined by $T(x,y,z) = (x+y,\ y+z)$.	L3	CO4	**05**
		OR			
	b.	Show that the subset $w = \{(x_1, x_2, x_3)/x_1 + x_2 + x_3 = 0\}$ of the vector space $V_3(R)$ is a subspace of $V_3(R)$.	L2	CO3	**05**
6.	**a.**	Find a least squares line for the given data points (2,3),(3,4),(4,3), (5,4),(6,3)and (7,4).	L3	CO5	**05**
		OR			
	b.	Reduce the matrix $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$ to the diagonal form. Hence find A^5.	L3	CO6	**05**
7.	**a.**	Show that $S = \{u_1, u_2, u_3\}$ is an orthonormal set, where $u_1 = \begin{bmatrix} \frac{3}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \\ \frac{1}{\sqrt{11}} \end{bmatrix}$, $u_2 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, u_3 = \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix}$.	L3	CO4	**05**
		OR			
	b.	Use python code to find intersection of three sets with suitable examples.	L3	CO2	**05**
		PART - C		**06x10=60**	
8.	**a.**	Three urns I, II and III contain respectively 2 white and 3 black balls,3 white and 4 black balls, and 4 white and 5 black balls. One of the urns is chosen randomly, the probability of each urn chosen being equal. Then one ball is drawn at random from the urn chosen and found to be black. Find is the probability that urn I, II or III was chosen.	L3	CO2	**10**
		OR			
	b.	Suppose 36% of families own a dog, 30% of families own a cat, and 22% of the families that have a dog also have a cat. A family is chosen at random and found to have a cat. Use python code to find the probability that they also own a dog.	L3	CO2	**10**
9.	**a.**	Twenty wrist watches in a box of 100 are defective. If ten watches are selected at random. Find the probability that **i.** 10 are defective **ii.** 10 are good **iii.** at least one watch is defective **iv.** at most 3 are defective.	L3	CO2	**10**
		OR			
	b.	A sample of 100 dry battery cells tested to find the length of life produced has a mean 12 hours and standard deviation 3 hours. Assuming the data to be normally distributed, Find the percentage of battery cells are expected to have life: **i.** More than 15 hours. **ii.** Less than 6 hours.	L3	CO2	**10**

		iii. between 10 and 14 hours.			
10.	**a.**	Show that the set V in a vector space over the field $F \in R$ mentioned under usual addition and scalar multiplication where $V = \left\{\begin{pmatrix} x & 0 \\ 0 & y \end{pmatrix}, x, y \in R\right\}$.	L2	CO3	**10**
		OR			
	b.	Find the four fundamental subspace of the matrix A Where $$A = \begin{pmatrix} 1 & -2 & -1 & 3 & 2 \\ 2 & -2 & -3 & 6 & 1 \\ -1 & -4 & 4 & -3 & 7 \end{pmatrix}$$	L3	CO3	**10**
11.	**a.**	Show that $\{u_1,\ u_2,\ u_3\}$ is an orthogonal set where $u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ $and\ u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$. Express $y = \begin{bmatrix} 6 \\ 1 \\ -8 \end{bmatrix}$ as linear combination of vector S.	L3	CO4	**10**
		OR			
	b.	Consider the basis $S = \{u_1,\ u_2,\ u_3\}$ for R^3 where $u_1 = (1,1,1)$, $u_2 = (-1,\ 0,\ -1)$ and $u_3 = (-1,\ 2,\ 3)$. Use the Gram- Schmidt process to transform S to orthonormal basis forR^3.	L3	CO5	**10**
12.	**a.**	The following data shows atmospheric pollutants y_i (relative to an EPA standard) at half hour intervals t_i. Find the quadratic equation to fit the data.	L3	CO5	**10**

t_i	1	1.5	2	2.5	3	3.5	4	4.5	5
y_i	-0.15	0.24	0.68	1.04	1.21	1.15	0.86	0.41	-0.08

		OR			
	b.	Find singular value decomposition of the matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$	L3	CO6	**10**
13.	**a.**	A random variable X has the following probability function for various values of x.	L2	CO2	**10**

x	0	1	2	3	4	5	6	7
$p(x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

i. Find $\boldsymbol{k}$

ii. Evaluate $\boldsymbol{P(X < 6)\ and\ P(3 < X \leq 6)}$.

iii. Also find the probability distribution and the distribution function of X.

		OR			
	b.	Use the Python code to visualize the following 3D vectors $$S = \{(1,0,0),(0,1,0),(0,0,1)\}$$	L3	CO3	**10**